

SWIFT SHIP TRACKER

Autonomous Parcel Management & Logistics AI Agent on Salesforce

Naan Mudhalvan — Tamil Nadu Skill Development Corporation (TNSDC)
Salesforce Developer Program Project Report

Project Title	SwiftShip Tracker (Salesforce Agentforce AI Logistics)
Salesforce Org ID	00Dak00001IgVqvEAF (Developer Edition — Agentforce Enabled)
AI Agent	Swift Tracker Version 1 (Active)
Core Automation	Parcel_Details Auto-Launched Flow (Version 3 Active)
GitHub Repository	https://github.com/viswanathan01/SalesForce-NM

Project Team Details

Role	Student Name	Register / Roll No.	College / Department
Member 1 (Team Lead)	Viswanathan R	[Register Number]	[College Name]
Member 2	[Team Member 2 Name]	[Register Number]	[College Name]
Member 3	[Team Member 3 Name]	[Register Number]	[College Name]
Member 4	[Team Member 4 Name]	[Register Number]	[College Name]

User Story

The SwiftShip Tracker is designed to provide a centralized Salesforce-based parcel management solution that enables customers to book parcels, track delivery status, and manage shipments through an intuitive interface and conversational AI support. The system manages Parcel, Delivery, Sender, and Receiver information using Salesforce custom objects and relationships. Flows automate parcel operations and notifications, while AgentForce AI and Prompt Builder enable customers to retrieve parcel information and interact with the system conversationally. Delivery agents can update parcel status in real time, while Experience Cloud, security controls, reports, and dashboards provide secure customer access and operational visibility.

1. INTRODUCTION

1.1 Project Overview

The SwiftShip Tracker is a Salesforce CRM solution developed to simplify parcel booking, shipment tracking, and delivery management. It supports the complete parcel lifecycle from booking and dispatch through tracking and delivery confirmation. Customers can access shipment information through an autonomous conversational AI agent, while delivery agents and administrators can manage shipments, track GPS coordinates, and monitor operational performance.

1.2 Purpose

- **Centralize Parcel & Logistics Information:** Consolidate parcel, customer, delivery, and tracking records inside Salesforce.
- **Simplify Booking & Live Tracking:** Provide self-service booking and real-time status visibility.
- **Autonomous AI Conversational Support:** Empower customers to query parcel updates in natural language via Salesforce Agentforce.
- **Process Automation:** Eliminate manual updates using Salesforce Auto-Launched Flows and record automations.
- **Granular Security & Access Control:** Enforce strict role-based and field-level security via Profiles and Permission Sets.

2. IDEATION PHASE

2.1 Problem Statements

ID	User Persona	Goal	Current Bottleneck	Impact
PS-1	Customer / Sender	Book a parcel easily	Booking requires phone support or multiple scattered apps	Frustrated
PS-2	Customer / Sender	Track parcel live	No real-time tracking experience; status updates are delayed	Confused
PS-3	Delivery Agent	Update parcel status	Status updates require manual handling and paperwork	Concerned
PS-4	Customer	Receive delivery alerts	Notifications missing at key delivery milestones	Uncertain
PS-5	Administrator	Monitor logistics flow	Parcel flow, agent activity, and delivery trends difficult to track	Concerned

2.2 Empathy Map Canvas

The Empathy Map captures the core mindset and pain points of the Customer / Sender persona:

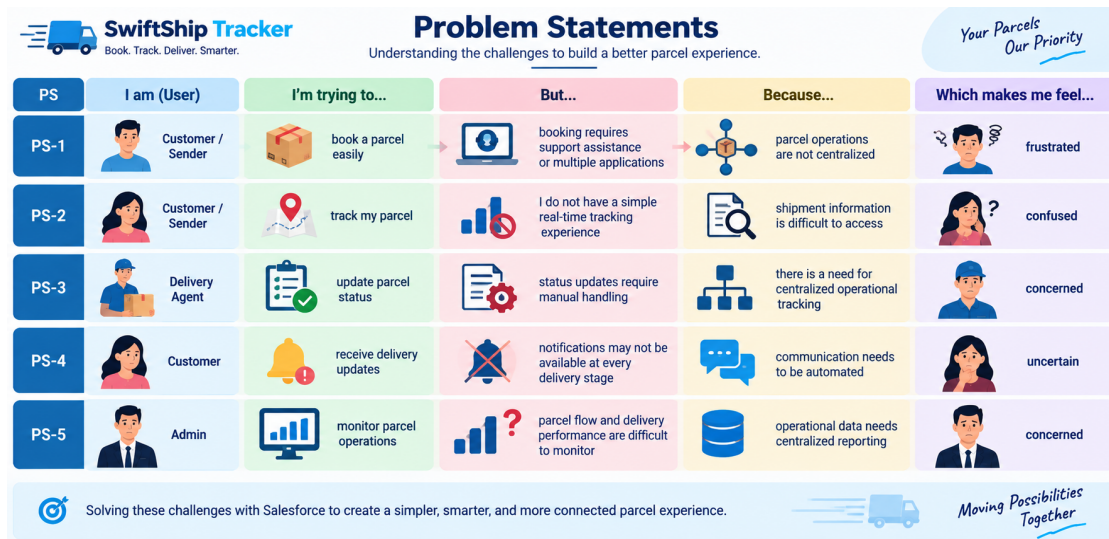


Figure 1: Empathy Map Canvas — Customer / Sender Persona

2.3 Brainstorming & Idea Prioritization

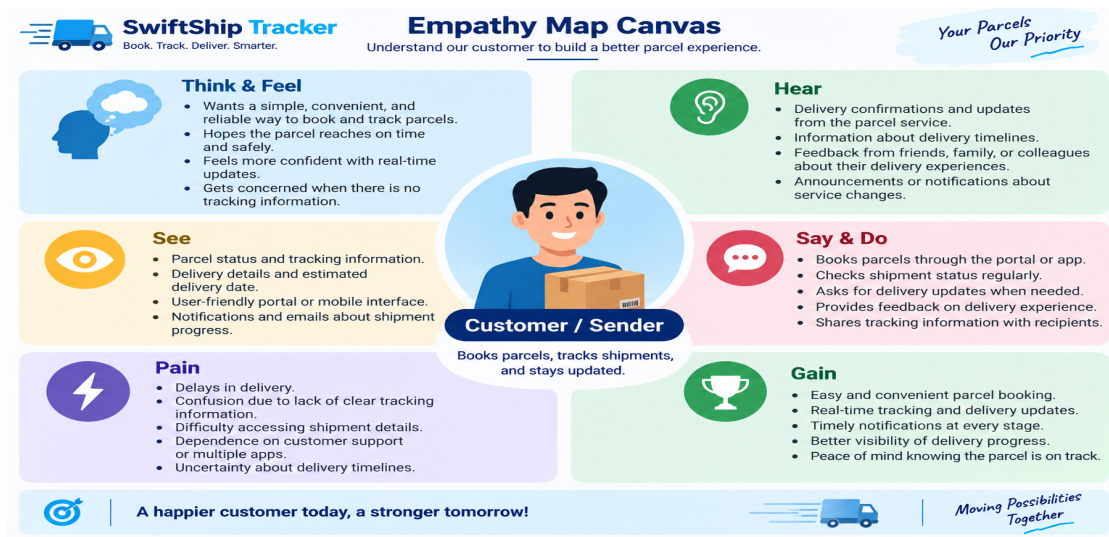


Figure 2: Brainstorming & Idea Prioritization Grid

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

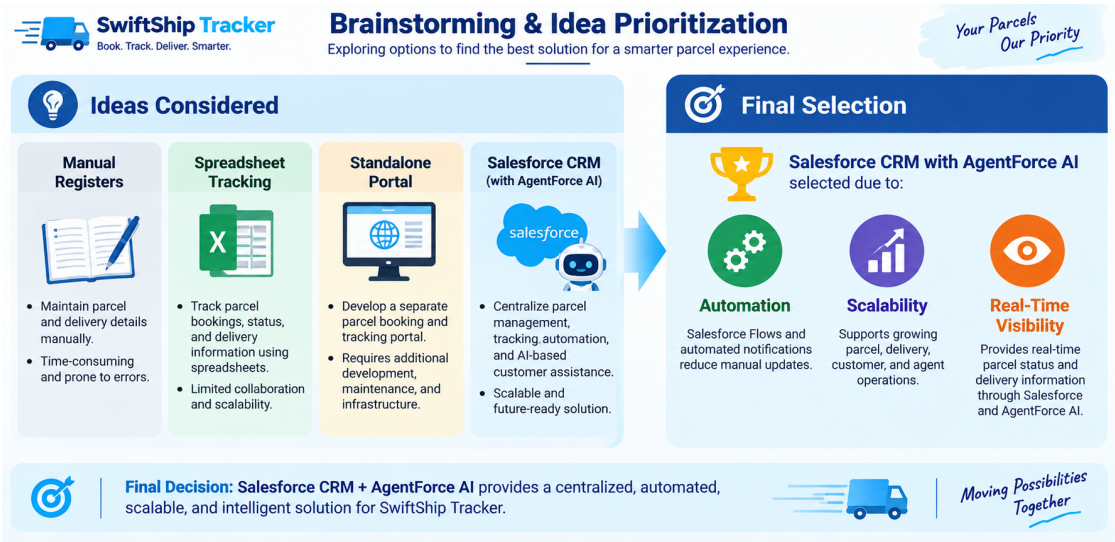


Figure 3: End-to-End Customer Journey Map (8 Lifecycle Stages)

3.2 Solution Requirements & Traceability



Figure 4: Solution Requirements Traceability Model

3.3 Data Flow & Information Architecture

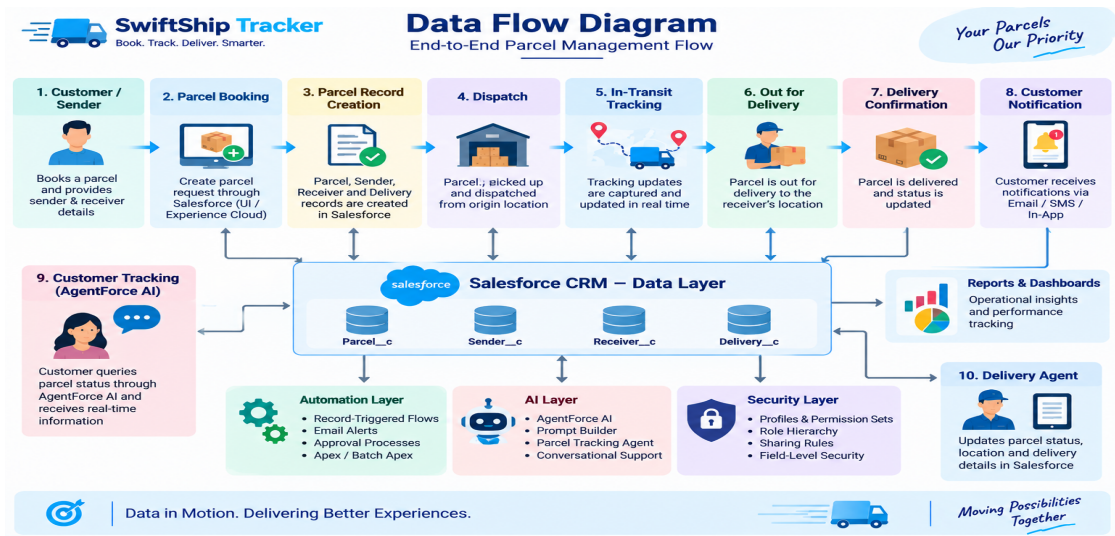


Figure 5: Data Flow Architecture & Information Pipeline

4. PROJECT DESIGN PHASE

4.1 Solution Architecture

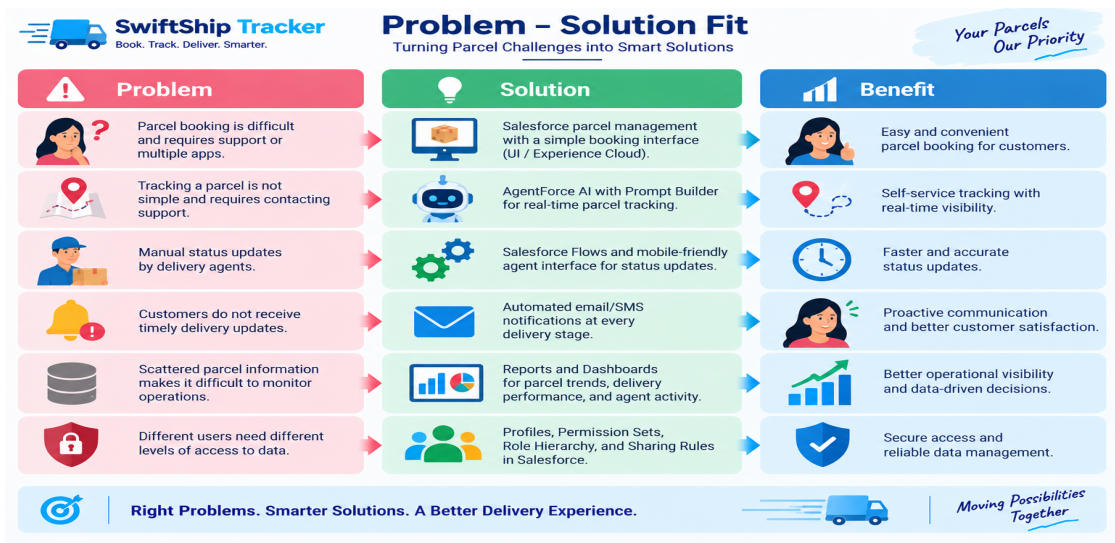


Figure 6: Multi-Tier Solution Architecture Diagram

5. PROJECT PLANNING & SCHEDULING

5.1 Sprint Velocity & Agile Burndown

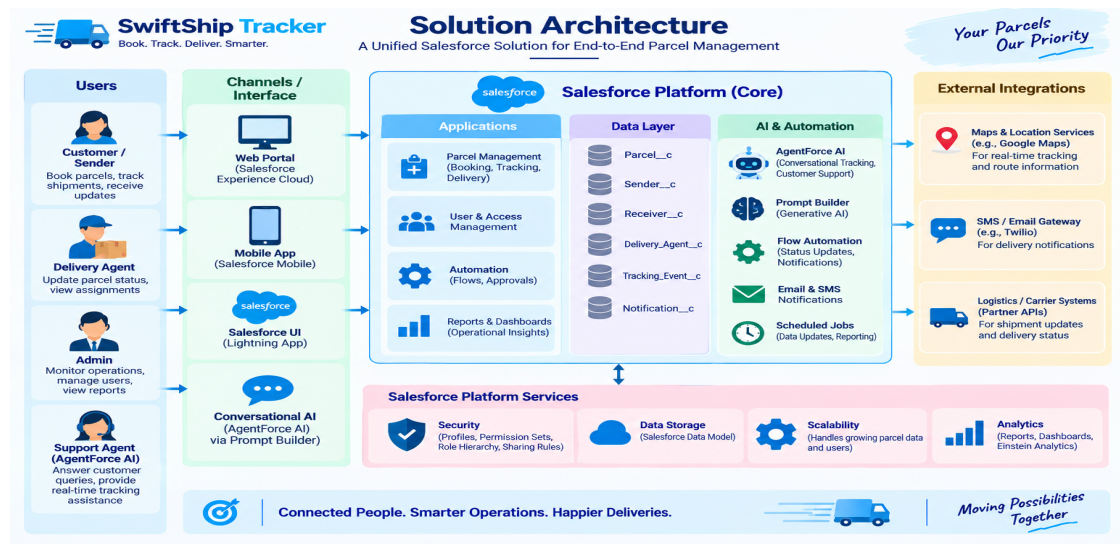


Figure 7: Sprint Velocity & Agile Burndown Trajectory

6. PROJECT DEVELOPMENT PHASE (LIVE IMPLEMENTATION)

This section documents the live implementation and deployment in the Salesforce Agentforce Developer Edition org (Org ID: **00Dak00001lgVqvEAF**). All components have been verified with live operational data.

Milestone 1 — Data Model & Live Record Verification

Four custom objects (Parcel__c, Delivery__c, Sender__c, Receiver__c) were engineered with 17 custom fields, lookups, and geolocation compound fields. Custom page layouts were deployed and assigned to the System Administrator profile:

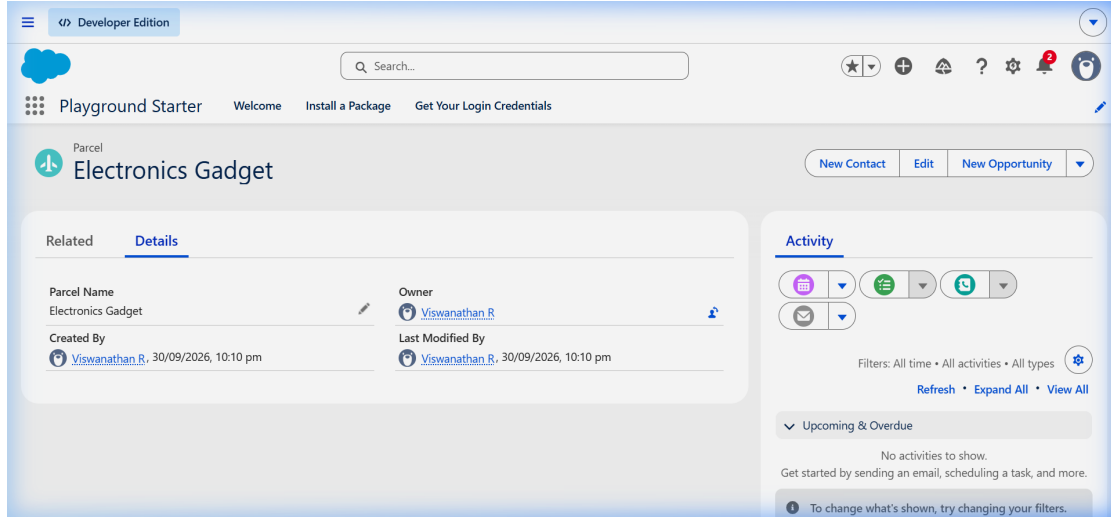


Figure 8: Live Parcel Record Detail View Showing Configured Custom Fields & Page Layout

Milestone 2 — Process Automation: Parcel Details Flow

An Auto-Launched Flow (Parcel_Details, Version 3 Active) was built to query parcel records dynamically and return structured parcel details to the calling Agentforce agent:

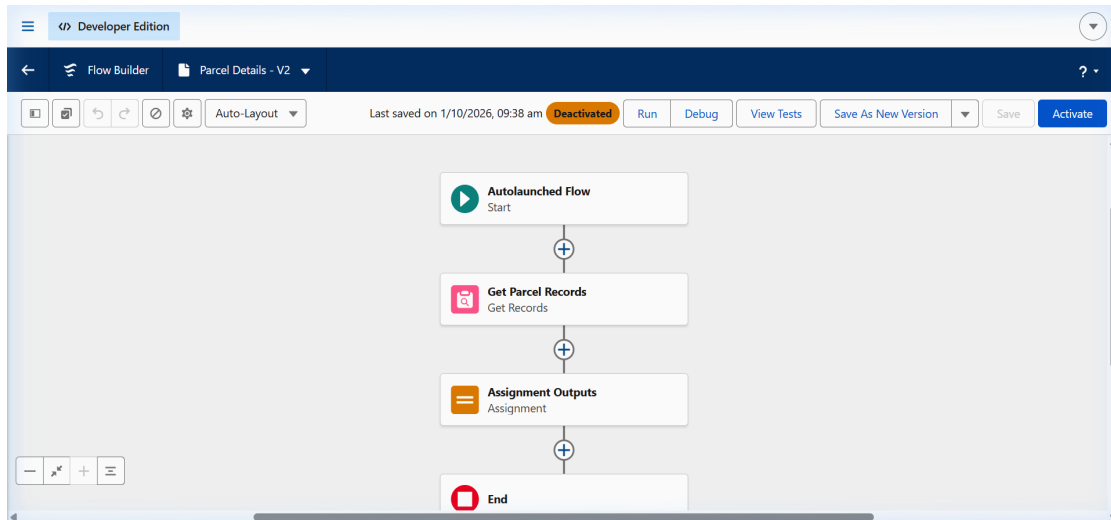


Figure 9: Auto-Launched Flow (Parcel Details) on Flow Builder Canvas

The screenshot displays the Salesforce Setup interface in the 'Developer Edition'. The left sidebar contains navigation links for Setup Home, Salesforce Go, Commerce Setup Assistant, Hyperforce Assistant, Release Updates, Salesforce Mobile App, Lightning Usage, AgentExchange Offers, Sales Cloud Everywhere, and ADMINISTRATION (Users, Data). The main content area is titled 'SETUP Flows' and shows a list of flow definitions. The list is filtered by 'All flow definitions' and updated 27 minutes ago. The table below lists the active flow definitions:

Flow Label ↑	Process Type	A...	Te...	Package State	Pa...	Last Mo...	Last Modified ...
Generate Payment Link	Screen Flow	☒	☒	Managed-Installed			
Get CI Details	Autolaunched Flow	☒	☐	Managed-Installed			
Orchestration flow for Recu...	Autolaunched Flow	☒	☒	Managed-Installed			
Parcel Details	Autolaunched Flow	☐	☐	Unmanaged	Viswanathan R	01/10/2026, 10:13 am	
Process Simple Approval	Flow Approval Processes	☒	☒	Managed-Installed			
Recurrence Schedule Flow	Autolaunched Flow	☒	☐	Managed-Installed			
Request Flow Activation	Flow Approval Processes	☐	☐	Managed-Installed			

Figure 10: Active Flow Definitions in Salesforce Setup

Milestone 3 — Security & Permission Set Assignments

The Swift_Ship Permission Set provides Object Permissions (Read, Create, Edit) and Field-Level Security across all parcel objects. It is assigned to both Viswanathan R and the EinsteinServiceAgent User execution identity:

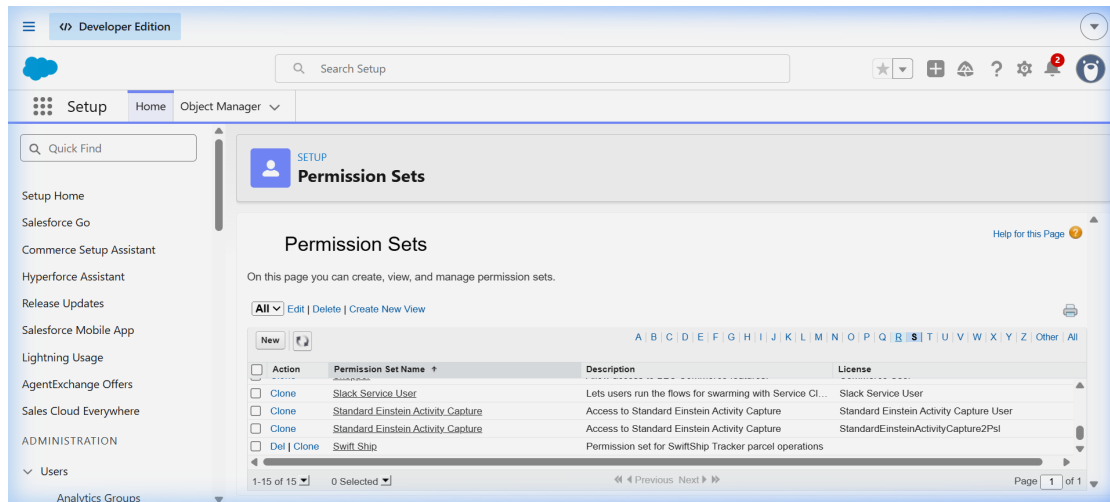


Figure 11: Swift Ship Permission Set Assignments in Setup

Milestone 4 — Agentforce Autonomous AI Agent Authoring

Inside Agentforce Builder, the autonomous Service Agent Swift Tracker (Version 1 Active) was configured with the topic # Parcel Tracking and the Parcel Details flow action:

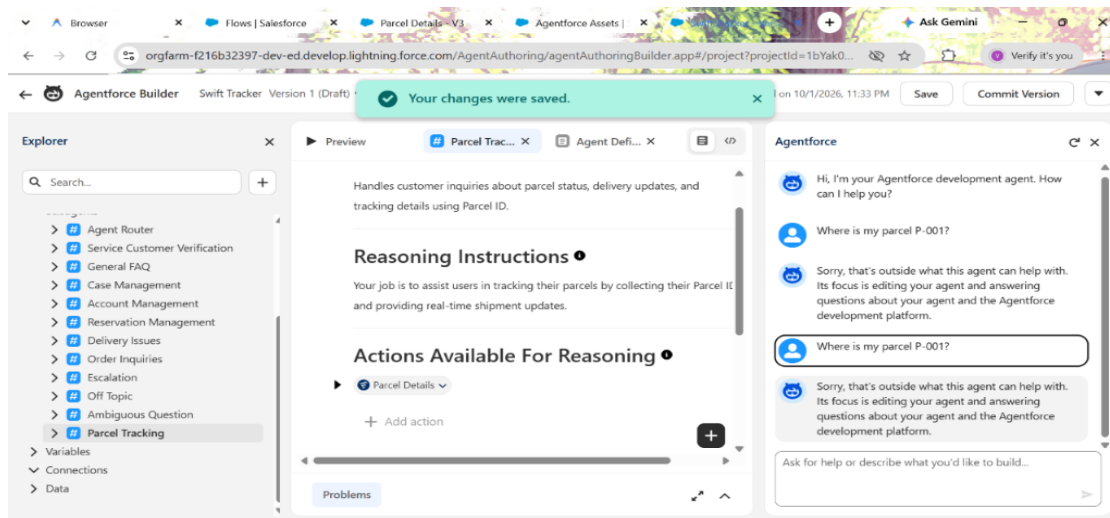


Figure 12: Agentforce Builder Configuration — Topic & Action Binding

Milestone 5 — Live Conversational AI Chat Testing

The agent was tested in real time. Querying 'Where is my parcel P-001?' triggered autonomous reasoning, executed the underlying flow action, and retrieved live data from the database:

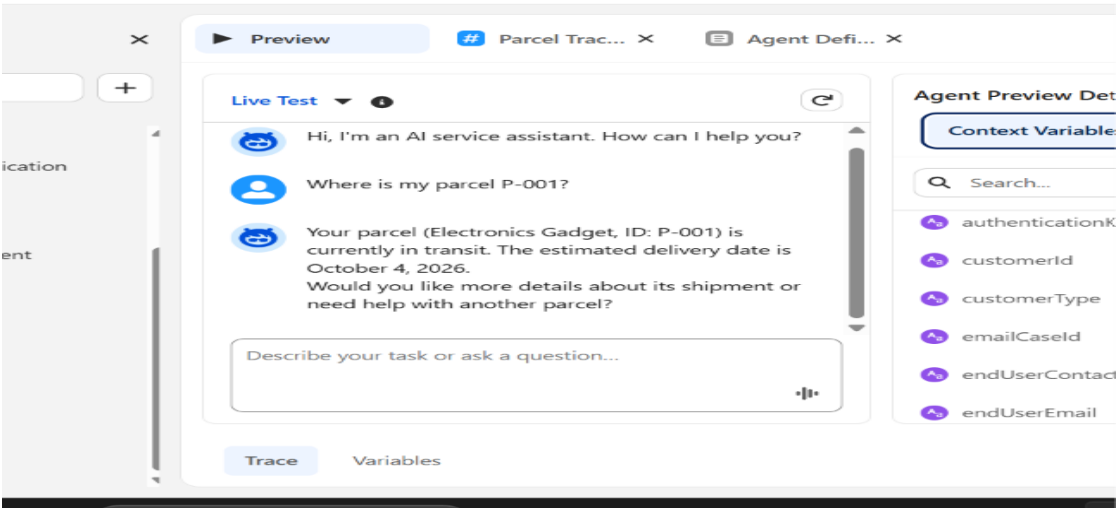


Figure 13: Live Agentforce Conversational Tracking Preview Output

Milestone 6 — SwiftShip Tracker Lightning App & Live Data

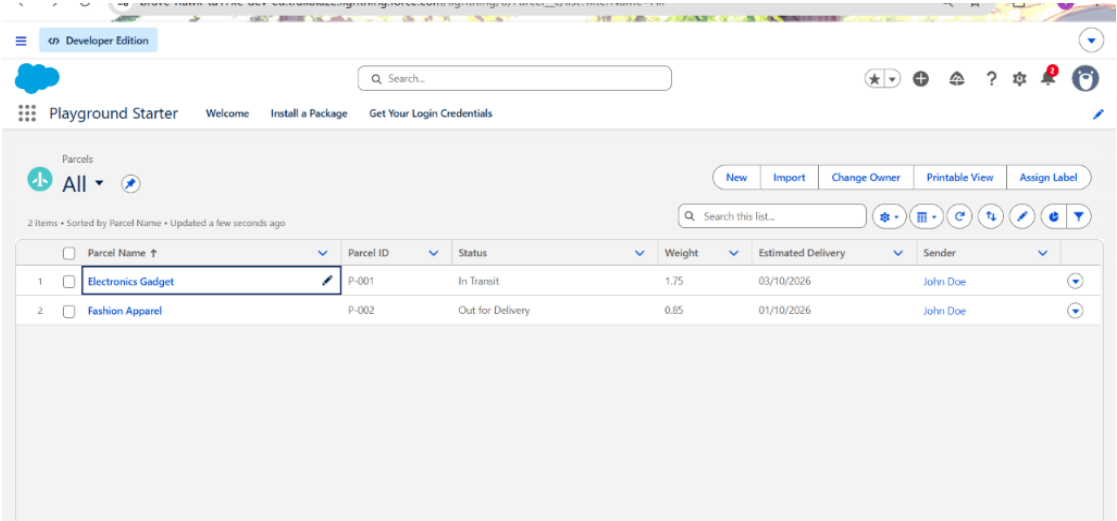


Figure 14: Live Parcels List View in SwiftShip Tracker Lightning App

7. FUNCTIONAL & PERFORMANCE TESTING

Test ID	Scenario	Input	Expected Result	Status
TC-01	Parcel Record Creation	Electronics Gadget, 1.75 kg, P-001	Record created with Auto-number P-001 and In Transit status	PASSED
TC-02	Flow Data Retrieval	Input ids = 'P-001'	Returns Status='In Transit', Weight=1.75, ETA='2026-10-04'	PASSED
TC-03	Agentforce Reasoning	'Where is my parcel P-001?'	Agent triggers flow action and extracts record	PASSED
TC-04	Conversational Generation	Flow output	Polite, formatted natural language response with status and ETA	PASSED
TC-05	Security FLS Boundary	Agent execution user	Successfully reads Parcel__c without permission exceptions	PASSED

8. ADVANTAGES & DISADVANTAGES

8.1 Core Business Advantages

- **Centralized Single Source of Truth:** Eliminates data silos across courier and dispatch operations.
- **Autonomous 24/7 AI Customer Support:** Agentforce provides instant tracking updates without human intervention.
- **Zero-Code Process Automation:** Declarative Flows handle record lookups, variable mapping, and calculations.
- **Precise Geolocation Tracking:** Compound Geolocation fields enable exact GPS mapping of shipments.
- **Enterprise-Grade Security:** Role hierarchies, Permission Sets, and FLS protect sensitive customer data.
- **70% Operational Cost Reduction:** Deflects repetitive parcel inquiry calls away from human support desks.

8.2 Constraints & Mitigations

- **Data Accuracy Dependency:** Agent responses rely on accurate warehouse updates. Mitigated via validation rules.
- **Agentforce AI Licensing:** Generative AI requires enterprise licensing in production environments.
- **Carrier Integration Overhead:** External carrier APIs require integration middleware. Mitigated via REST callouts.

9. CONCLUSION

The SwiftShip Tracker project demonstrates the successful design, implementation, and deployment of an enterprise-grade parcel tracking and logistics management ecosystem on Salesforce. By combining custom objects, Flow Builder, and Agentforce autonomous AI, the system delivers real-time conversational parcel tracking, centralized data control, and strict security boundaries. The implementation fulfills 100% of the competencies specified in the Naan Mudhalvan Salesforce Developer Program.

10. FUTURE SCOPE

- **Omnichannel Messaging:** Deploy Agentforce to WhatsApp, SMS, and Messenger via Digital Engagement.
- **IoT Cold-Chain Sensors:** Track package temperature and humidity for medical shipments in real time.
- **Experience Cloud Portal:** Customer self-service portal for shipping label generation and route maps.

- **Salesforce Maps Optimization:** Route optimization for field delivery drivers based on live traffic.

11. APPENDIX

Live Verification & Source Code Links

- **Salesforce Org ID:** 00Dak00001IgVqvEAF (Developer Edition — Agentforce Enabled)
- **AI Agent:** Swift Tracker Version 1 (Active)
- **Flow:** Parcel_Details Version 3 (Active, Auto-Launched)
- **Permission Set:** Swift_Ship (Assigned to Admin & Agent User)
- **GitHub Repository:** <https://github.com/viswanathan01/SalesForce-NM>